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Explainable ML for Flood Prediction in Bangladesh

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Abstract*—Floods are a recurring and devastating hazard in Bangladesh, causing extensive socio-economic losses. This study proposes an explainable machine learning framework for regional flood prediction using multiyear meteorological and geospatial variables. We evaluate multiple regression models, including linear, support-vector, decision-tree, and ensemble methods, with Gradient Boosting emerging as the most accurate (R^2 = 0.964, RMSE = 0.0667) in this study. Dimensionality reduction via PCA confirmed the robustness of tree-based ensembles under compressed feature spaces. To enhance transparency, we apply SHAP and LIME, which consistently identify rainfall, humidity, and cloud cover as dominant predictors, aligning with hydrometeorological reasoning. Unlike prior binary flood/no-flood approaches, this work treats flood intensity as a continuous target, enabling finer-grained risk estimation. The findings demonstrate that interpretable ensemble models can provide reliable and actionable flood forecasts, supporting early warning and climate-resilient planning in Bangladesh.

***Keywords*— Flood prediction, Machine learning, Regression, Explainable AI, PCA, SHAP, LIME**

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I. Introduction

Floods are among the most destructive natural hazards, causing severe economic, social, and environmental losses worldwide. Global flood damage exceeded $19 billion in 2012 and continues to increase due to climate change and rapid urbanization [1]. Bangladesh is particularly vulnerable: since 1972, the country has experienced 297 major natural disasters, with floods and storms accounting for nearly 75% of all events and over 42,000 flood-related fatalities [2]. Accurate and timely flood forecasting is therefore critical for evacuation planning, resource allocation, and disaster-risk reduction.

Traditional hydrological and hydraulic models provide physical realism but require dense sensor networks, extensive calibration, and high computational cost, limiting their use in data-scarce or operational environments [3], [4]. Machine learning (ML) offers a scalable alternative by learning nonlinear relationships directly from historical data. However, much existing work focuses on flood *classification*, producing only categorical outputs that lack severity information.

This study presents an explainable supervised ML framework for continuous flood-intensity regression using multiyear meteorological and geospatial data from Bangladesh. The methodology combines leakage-aware validation, principal component analysis (PCA) for collinearity control, and benchmarking of multiple regressors using MAE, MSE, RMSE, and R^2 [5]. Transparency is ensured through SHAP (global) and LIME (local), revealing how rainfall, temperature, humidity, cloud coverage, and spatial features influence predictions.

The main contributions are: (i) a leakage-safe time–space-aware evaluation framework for flood regression; (ii) ablation of station identity encoding and PCA; (iii) explainable ML with stability and extreme-event analysis.

The paper is organized as follows: Section II reviews related work; Section III describes data, methods, and evaluation; Section IV presents results; Section V discusses implications and limitations; and Section VI concludes. .

II. Related Work

Flood prediction has been broadly studied using statistical models, physics-based hydrological and hydraulic approaches, and machine learning (ML) techniques. Physics-based models provide strong physical interpretability through rainfall–runoff simulation and river routing; however, they often require dense observational networks, extensive calibration, and high computational cost, limiting applicability in data-scarce environments such as Bangladesh [6]. By contrast, ML-based approaches offer flexible, data-driven alternatives capable of learning nonlinear relationships between meteorological drivers and flood response without explicit physical modeling.

Tree-based ensemble methods are consistently reported to outperform linear models in rainfall–runoff modeling and flood prediction, often achieving $R^2 > 0.90$ across multiple case studies [7], [8]. Gradient-boosting architectures, including XGBoost and LightGBM, further improve predictive accuracy by sequentially correcting residual errors and handling heterogeneous feature interactions effectively [7].

A large portion of flood research formulates the task as classification, focusing on flood vs. no-flood labeling or categorical severity levels. Ensemble classifiers such as Random Forest and boosting methods frequently report AUC values above 0.95 in varied climatic conditions [9], [10]. Using the same Bangladesh hydrometeorological dataset analyzed

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in this work, Gauhar *et al.* applied correlation-based feature selection and k-NN classification, achieving over 94% accuracy, illustrating the effectiveness of classification methods for discrete flood risk assessment. However, such classification-based frameworks do not provide continuous severity estimates required for operational decision-making and early intervention.

Recent studies have integrated deep learning and remote sensing for flood detection. For example, fog-enabled early-warning systems combining LSTM networks, Random Forest classifiers, and Sentinel imagery have demonstrated improved flood occurrence prediction using real-time sensor data and optimized network infrastructures [11]. In parallel, deep spatiotemporal architectures such as CNN–LSTM hybrids and U-Net variants have achieved strong performance on gridded datasets, reporting Nash–Sutcliffe efficiency exceeding 0.80 and pixel-level AUC near 0.96 [12], [13]. Nevertheless, such models typically depend on dense spatiotemporal inputs, large labeled datasets, or complex spatial representations, which limits feasibility in station-level forecasting systems.

Unlike most previous studies that emphasize categorical flood detection, this work formulates flood prediction as a supervised regression problem to estimate continuous flood intensity. This enables more precise risk quantification beyond binary alerts. Multiple regression models are evaluated on multiyear meteorological and geospatial data, with Gradient Boosting emerging as the best-performing approach. Additionally, SHAP and LIME are employed for explainability, offering transparent insight into feature contributions. These aspects distinguish the proposed framework from classification-driven literature and provide a scalable, interpretable alternative for flood prediction in data-limited environments.

III. Methodology

The primary objective of this research is to minimize prediction error in flood forecasting by developing and evaluating supervised machine learning models trained on multiyear hydrometeorological and geospatial data. Figure 1 provides an outline of the research architecture, while detailed insights into the study are presented in sections A through G.

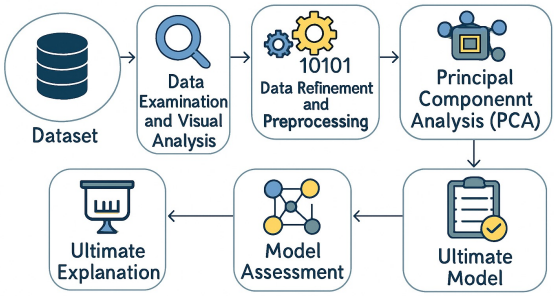

Fig. 1. Architecture of the proposed framework.

A. dataset

This study uses a tabular hydrometeorological dataset comprising 20,544 records and 19 columns at station–time resolution [14], [15]. Features span station identifiers/geolocation and meteorological variables (e.g., Rainfall, Relative_Humidity, Wind_Speed, Cloud_Coverage, Max_Temp, Min_Temp), with *Flood?* as the continuous target; Table I lists all fields, units, and brief descriptions.

TABLE I
Descriptive Features of the Dataset

Features	Description	Type	Unit
Sl	Serial number of record	Num	–
Station Names	Meteorological station name	Cat	–
Year	Calendar year of observation	Num	Year
Month	Month of observation	Num	Month
Max Temp	Daily max temperature	Num	(°C)
Min Temp	Daily min temperature	Num	(°C)
Rainfall	Total daily rainfall	Num	mm
Relative Humidity	Avg daily relative humidity	Num	%
Wind$_{speed}$	Avg daily wind speed	Num	m/s
Cloud Coverage	Avg daily cloud cover	Num	Oktas or %
Bright Sunshine	Daily sunshine duration	Num	Hours
Station Number	Official station ID	Num	–
X COR	Projected X coordinate	Num	m
Y COR	Projected Y coordinate	Num	m
LATITUDE	Geographic latitude	Num	degrees
LONGITUDE	Geographic longitude	Num	degrees
ALT	Station altitude	Num	m
Period	Observation period code/group	Cat	–
Flood?	Flood indicator/score (target)	Num	–

B. Data Examination and Visual Analysis

Exploratory visual analysis was performed using histograms, density plots, box plots, and correlation heatmaps in order to understand feature distributions, detect outliers, and identify potential multicollinearity prior to modeling. These visual diagnostics supported preprocessing decisions such as normalization, outlier handling, and dimensionality reduction.

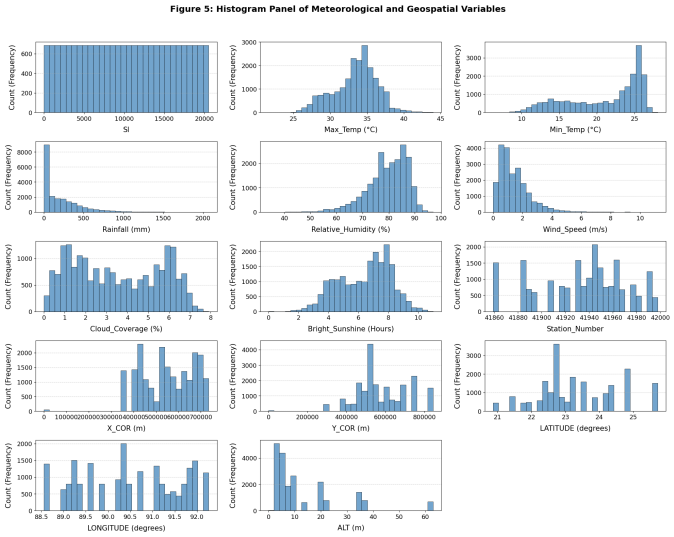

Fig. 2. Histograms showing feature distributions across stations.

Histograms (Fig. 2) reveal important distributional characteristics of the hydrometeorological variables. Rainfall exhibits a highly skewed, zero-inflated distribution with a long right tail, reflecting many dry days interspersed with extreme events. Temperature variables display approximately unimodal distributions with seasonal spread, while Relative_Humidity, Wind_Speed, and Cloud_Coverage show moderate dispersion. These patterns informed transformation strategies and model selection under non-Gaussian conditions.

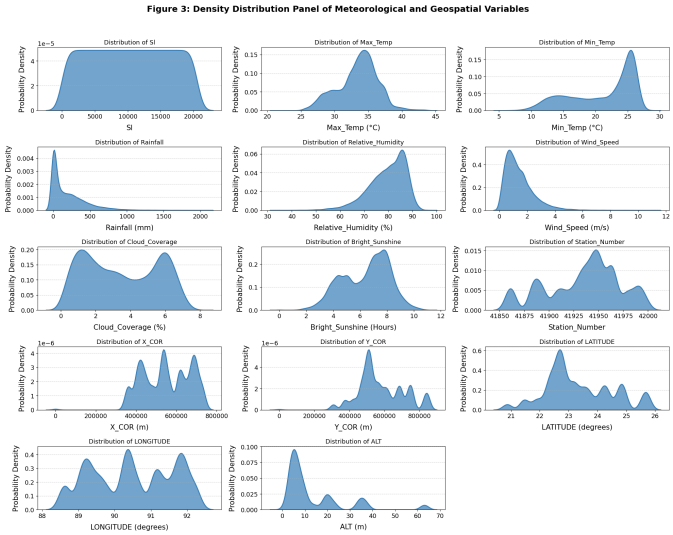

Fig. 3. Kernel density estimates of input variables.

Density plots (Fig. 3) further illustrate probability structure and modality. Rainfall remains sharply concentrated near zero, while temperature and humidity demonstrate stable central tendencies. Spatial features (ALT, LATITUDE, LONGITUDE) exhibit clustered distributions consistent with station locations rather than continuous variation, justifying later station-based encoding.

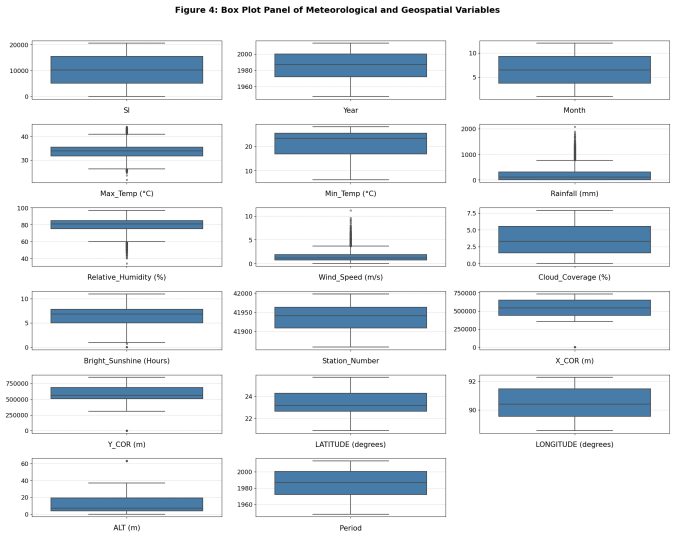

Fig. 4. Box plots highlighting outliers and variability.

Box plots (Fig. 4) summarize variability and outliers. Rainfall presents frequent extreme values, while temperature and humidity show lower dispersion. Wind_Speed and Cloud_Coverage display moderate spread with episodic peaks, consistent with storm-driven conditions.

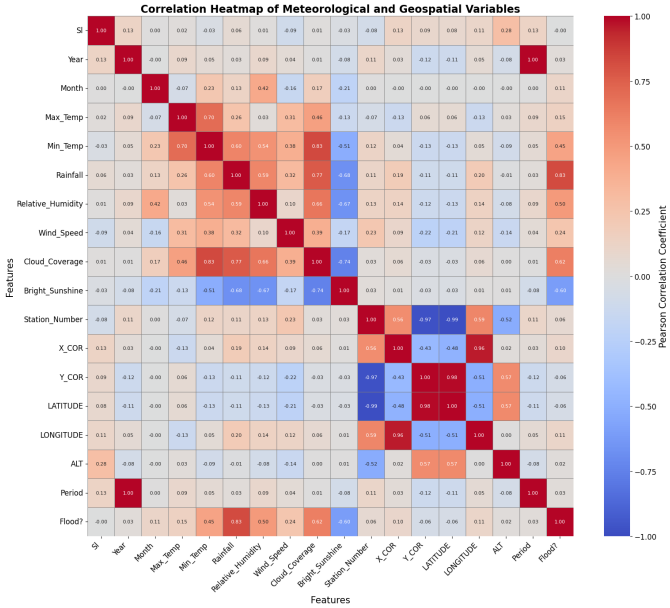

Fig. 5. Correlation heatmap of hydrometeorological and spatial features.

The correlation heatmap (Fig. 5) visualizes linear dependencies among features. Strong correlations appear between Max_Temp and Min_Temp, and negative associations emerge between Bright_Sunshine and Cloud_Coverage. These relationships indicate multicollinearity among climate variables and motivate dimensionality reduction using PCA and model selection favoring ensemble methods robust to correlated inputs.

C. Data Refinement and Preprocessing

During preprocessing, no missing values were found; therefore, no imputation was required. As `Station_Names` is a categorical variable, one-hot encoding was applied to create binary indicator features, avoiding artificial ordinality and ensuring compatibility with regression models [16]. Station identifiers were encoded to preserve station-specific climatological regimes that are not fully described by latitude, longitude, or elevation alone. Bangladesh exhibits strong localized variability driven by monsoon systems, river-basin dynamics, and urban heat effects; therefore, station identity serves as an implicit proxy for hidden geophysical influences. Although one-hot encoding increases feature dimensionality, ensemble tree-based models are robust to sparse categorical representations through implicit feature selection and regularization. This preserves spatial heterogeneity while integrating smoothly with continuous predictors such as rainfall, humidity, temperature, wind speed, cloud coverage, and sunshine duration. Continuous variables were normalized to promote

numerical stability and comparable feature influence across models.

D. Ensemble Model Regression Construction

The study compares linear and ensemble regressors—Linear [17] and Ridge Regression, Support Vector Regression (SVR) [18], Decision Tree [19], Random Forest Regressor (RFR) [20], Gradient Boosting Regressor (GBR) [21], and XGBoost (when included)—for flood prediction from hydrometeorological and geospatial features. Across both result sets, tree-based ensembles dominate, with RFR and GBR consistently achieving the lowest errors and highest variance explained.The evaluation revealed the Gradient Boosting Regressor (GBR) ensemble method as the most effective for this assessment. while linear baselines trail and SVR sits between linear and ensembles depending on the split. **Additional deep-learning experiments** We conducted exploratory tests using LSTM (station-wise sequences), ConvLSTM (gridded station interpolation), and a GNN (stations as k-NN graph nodes) with identical inputs as the tabular models and chronological splits. As none outperformed ensemble regressors, detailed results are omitted.

E. Model Assessment

The metrics MSE, MAE, RMSE, and R² are used in the context of regression to evaluate the performance of all models [16] .Table II illustrates the measurement of different performance metrics.These metrics aid in determining the prediction and generalization ability of a regression model.In general, the goal is to maximize R² while minimizing MSE, MAE, and RMSE.

TABLE II
IN-DEPTH OVERVIEW OF PERFORMANCE METRICS

Name	Equation	Meaning
MSE	$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$	Average squared error; lower values indicate better predictive performance.
MAE	$\frac{1}{n}\sum_{i=1}^{n}\lvert Y_i - \hat{Y}_i\rvert$	Average absolute error; robust to outliers compared with MSE.
RMSE	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}$	Square-root of MSE; in the same units as the target and penalizes large errors.
R^2	$1 - \frac{\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$	Proportion of variance explained; closer to 1 indicates better fit.

F. Ultimate Model

From the comparative evaluation of all trained regressors using MAE, MSE, RMSE, and R², Gradient Boosting Regressor emerged as the top-performing model on this dataset; therefore, Gradient Boosting is selected as the final model for flood prediction. This choice reflects its consistently low errors and high variance explained relative to linear baselines, SVR, Decision Tree, Random Forest, and XGBoost across the reported splits.

G. Outcome Explanation

This study employs explainable AI (XAI) techniques to analyze feature influence in the best-performing model, improving transparency and trust in the decision-making process. XAI methods provide human-interpretable insights into model behavior and help identify potential biases. We apply SHAP and LIME to explain both global and local model behavior. SHAP quantifies each feature's contribution to predictions, enabling overall importance ranking, while LIME provides instance-level explanations through locally interpretable approximations. Together, these methods ensure that the flood-intensity predictions are both reliable and understandable.

IV. RESULTS

Table III presents the overall performance of all models on the flood dataset using MAE, MSE, RMSE, and R^2 for comparison. Across both result tables, the Gradient Boosting Regressor and Random Forest Regressor achieve the strongest outcomes, with Gradient Boosting reaching as low as MAE 0.03307, MSE 0.00445, RMSE 0.0667, $R^2 = 0.9639$ in one split and Random Forest reaching MAE 0.03035, MSE 0.00561, RMSE 0.0749, $R^2 = 0.9545$ in another, while consistently outperforming linear baselines and SVR on error magnitude and variance explained. In the complementary split, Random Forest again leads the cohort (MAE 0.04933, MSE 0.01176, RMSE 0.1084, $R^2 = 0.9047$), with Gradient Boosting and XGBoost close behind, confirming the robustness of tree-based ensembles across partitions of the data. This comparative evidence indicates that flood prediction with this feature set is particularly well-suited to boosted and bagged decision-tree ensembles, validating their effectiveness in predictive accuracy and generalization under standard regression metrics.We also evaluated several spatio-temporal deep-learning approaches (LSTM, ConvLSTM, and GNN) in preliminary experiments; however, these architectures did not outperform the gradient-boosted ensembles on our station-level tabular dataset and are therefore not included in the main performance table.

TABLE III
PERFORMANCE METRICS FOR VARIOUS MODELS

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.1264	0.02999	0.1731	0.7568
Ridge Regression	0.1264	0.02999	0.1731	0.7568
Support Vector Regression	0.0900	0.02000	0.1414	0.8600
Random Forest Regressor	**0.03035**	0.00561	0.0749	0.9545
Gradient Boosting Regressor	0.03307	**0.00445**	**0.0667**	**0.9639**
Decision Tree Regressor	0.04418	0.01241	0.1114	0.8993

The fold-wise performance analysis shows Fold 1 as the best, with the lowest MSE (0.0036) and highest R^2 (0.9714). Fold 9 performs worst, with the highest MSE (0.0107), lowest R^2 (0.9227), and highest MAE (0.0449). Overall, the model

demonstrates strong robustness and consistent accuracy across all folds, confirming reliable predictive performance.

TABLE IV
10-FOLD CROSS-VALIDATION RESULTS (SELECTED MODEL)

Fold	MAE	MSE	R^2
Fold 1	0.0316	0.0036	0.9714
Fold 2	0.0321	0.0041	0.9659
Fold 3	0.0425	0.0101	0.9289
Fold 4	0.0430	0.0098	0.9317
Fold 5	0.0422	0.0094	0.9328
Fold 6	0.0452	0.0105	0.9243
Fold 7	0.0426	0.0105	0.9264
Fold 8	0.0453	0.0109	0.9232
Fold 9	0.0449	0.0107	0.9227
Fold 10	0.0415	0.0090	0.9352

Post-PCA, Random Forest and XGBoost yield the best outcomes, with Random Forest showing the lowest errors and highest explained variance (MAE 0.0493, R^2 0.9047). XGBoost follows closely (MAE 0.0564, R^2 0.8931). Support Vector Regression remains competitive (MAE 0.1038, R^2 0.8034), while linear models lag in performance, highlighting the advantages of tree-based ensembles.

TABLE V
REGRESSION PERFORMANCE AFTER PCA (34 COMPONENTS)

Model	MAE	MSE	RMSE	R^2
Linear Regression	0.1820	0.0501	0.2239	0.5934
Ridge Regression	0.1820	0.0501	0.2239	0.5934
Lasso Regression	0.1987	0.0671	0.2590	0.4559
Support Vector Regression	0.1038	0.0242	0.1557	0.8034
Random Forest Regression	**0.0493**	**0.0118**	**0.1084**	**0.9047**
Gradient Boosting Regression	0.0864	0.0228	0.1511	0.8148
XGBoost Regression	0.0564	0.0132	0.1148	0.8931

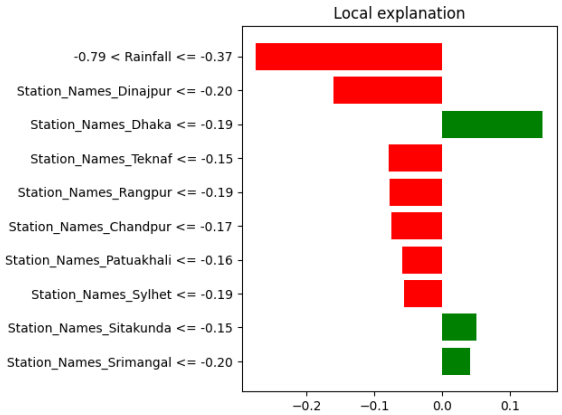

Fig. 6. LIME analysis with key features influencing an individual prediction.

Figure 6 presents the LIME explanation for a specific instance, providing local interpretability for the Gradient Boosting Regressor. For the explained case, Rainfall (value range -0.79 to -0.37) shows a notable negative contribution, whereas Station_Names_Dhaka and Station_Names_Srimangal contribute positively. This clarifies why the model produced the given prediction for that point.

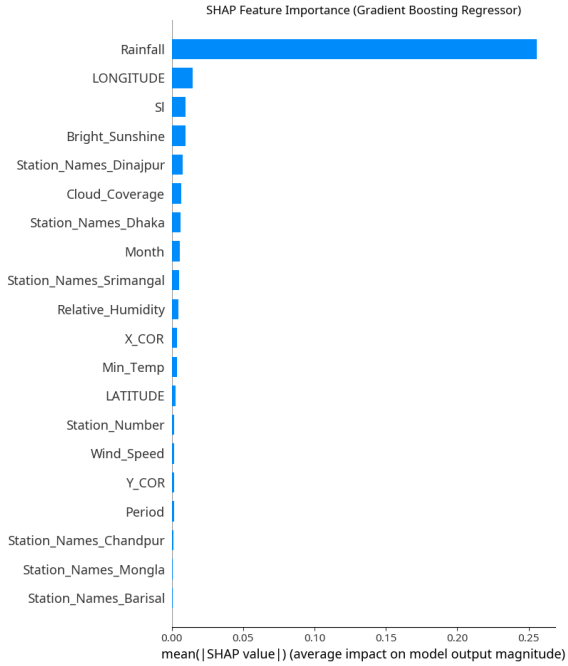

Fig. 7. SHAP summary: feature impacts for the Gradient Boosting Regressor.

Figure 7 shows the SHAP plot for the final model, the Gradient Boosting Regressor. The feature Rainfall emerges as a key influencer, while LONGITUDE and Sl also contribute. This visualization summarizes average impact magnitudes and directions, enhancing understanding of model behavior and feature contributions.

V. DISCUSSION

The Gradient Boosting Regressor excelled in capturing nonlinear relationships in hydrometeorological data, achieving the lowest MAE (0.03307) and highest R2 (0.9639). Ensemble models like Random Forest maintained performance with PCA reduction. Rainfall was the top predictor, supported by cloud cover and humidity. Trust in model outputs was bolstered by LIME and SHAP explanations, confirming its potential for flood prediction in areas like Bangladesh.

This research does show, nevertheless, that the management of flood risk can be informed by data-driven, effective, and interpretable models. Future research should emphasize real-time data, spatiotemporal modeling, and socioeconomic exposure to promote accuracy, policy relevance, and policy impact. A key limitation revealed by our exploratory deep-learning experiments is data density and format. ConvLSTM and GNN architectures require either dense gridded inputs or explicit hydrological connectivity (river topology) to fully exploit spatial dependencies; when trained on interpolated grids or k-NN topology derived from station coordinates, these models underperformed compared to tree-based ensembles. This suggests that, for operational Bangladesh datasets that are station-based and not densely gridded, explainable ensemble methods

provide a better trade-off between accuracy, robustness, and interpretability. Future work will explore hybrid data sources (satellite-derived rainfall fields, discharge measurements, and river-network edges) to more fairly evaluate deep spatio-temporal models.

VI. Conclusion

Bangladesh is highly vulnerable to flooding, making accurate and trustworthy prediction systems essential for disaster preparedness and mitigation. In this study, we developed an explainable machine learning framework for flood-intensity regression using multi-year meteorological and geospatial observations from Bangladesh. Among the evaluated models, the **Gradient Boosting Regressor** consistently achieved the strongest predictive performance, demonstrating that ensemble learning is highly effective for station-level flood modeling.

To ensure model transparency and usability in risk-sensitive applications, we incorporated SHAP and LIME for explainability. Both methods consistently identified rainfall as the most influential predictor, followed by geospatial and topographic attributes such as longitude and slope. These results provide actionable insights into the environmental drivers of flooding and increase confidence in the model's predictions for deployment in operational settings.

In addition to ensemble models, we conducted preliminary experiments with deep-learning architectures including LSTM, ConvLSTM, and Graph Neural Networks. However, under the current station-level data setting, these approaches did not outperform gradient-boosted ensembles. This highlights that, in data-limited and non-gridded environments such as Bangladesh's monitoring network, interpretable tree-based methods remain a strong and practical choice over more complex deep models.

In future work, we aim to:

- integrate the proposed framework into real-time early-warning systems,
- incorporate river discharge, soil moisture, and satellite-based rainfall products to better capture hydrological dynamics,
- and re-evaluate deep spatio-temporal models using richer gridded and sequential inputs.

Overall, this study demonstrates that explainable ensemble learning can outperform complex architectures on operational flood datasets while remaining transparent and reliable. The proposed framework offers a realistic and scalable solution for flood risk assessment in Bangladesh and similar data-constrained regions.

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